

Appendix A - Abrolhos Marine Park Western Rock Lobster Report

Yamatji Shallow Bank, 2025 and 2026 pot surveys

Table of contents

Survey effort	1
Pot locations	2
Abundance	2
Size structure	5
Standardised size distributions	9
Modelling: catch inside vs outside the National Park Zone	11
Legal males and females	16
Data notes	17

Survey effort

Pots were set on Yamatji Shallow Bank in April 2025 and April 2026, across the Abrolhos Marine Park National Park Zone (no take) and Special Purpose Zone (open to commercial and recreational fishing).

Table 1: Pots deployed and lobsters measured, by year and zone

Year	Zone	Pots deployed	Pots counted	Lobsters measured
2025	National Park Zone	55	54	273
2025	Special Purpose Zone	93	91	483
2026	National Park Zone	55	53	124
2026	Special Purpose Zone	84	83	248

Set times were not recorded in 2026, so soak time is available for 2025 only and catch is reported per pot rather than per pot-hour.

Pot locations

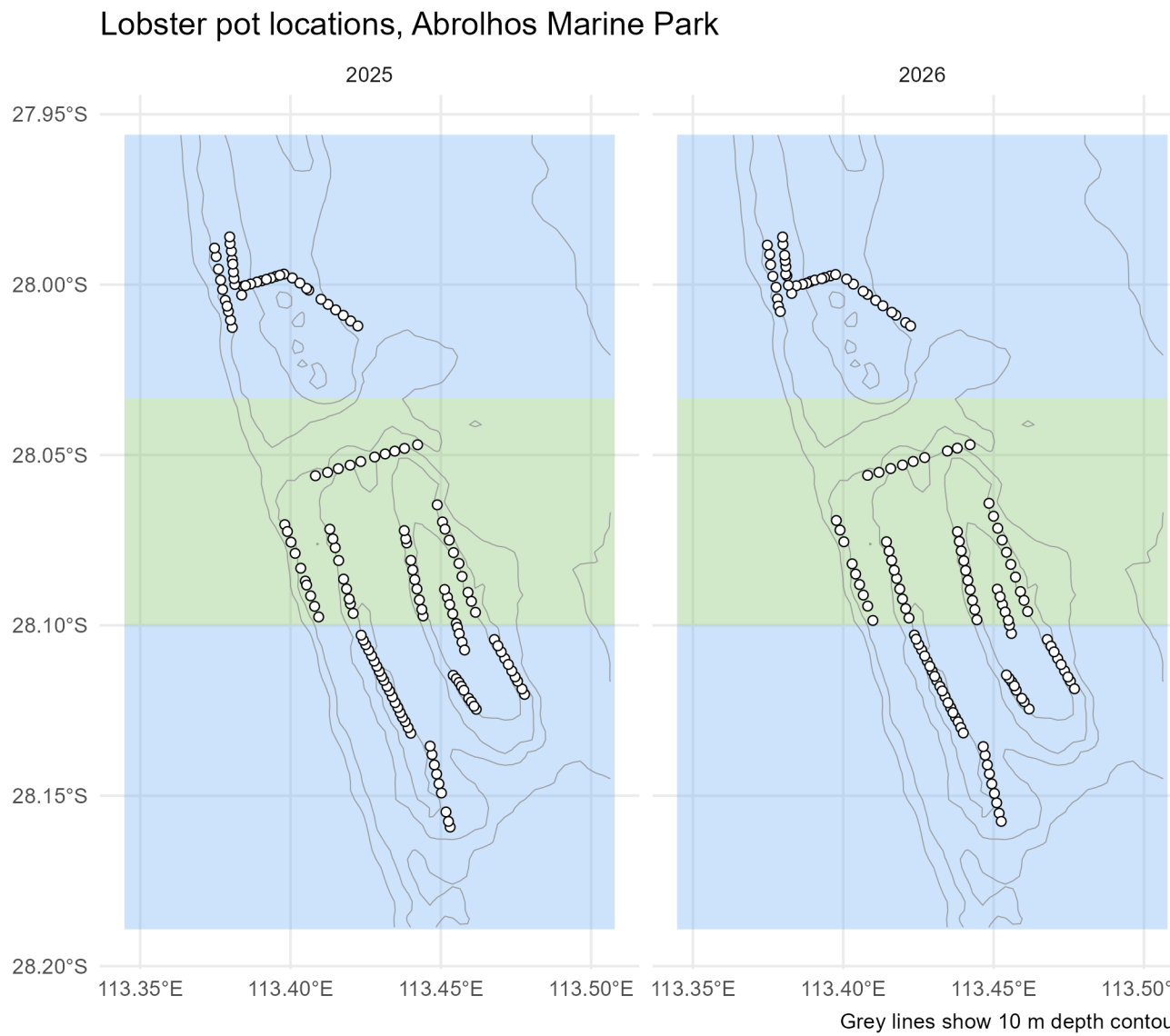


Figure 1: Pot locations in each survey year, over marine park zones and 10 m depth contours

Abundance

Table 2: Catch rate per pot (mean and standard error), by year and zone

Year	Zone	Pots	Total lobsters	Mean per pot	SE	Mean legal per pot	SE legal
2025	National Park Zone	54	261	4.83	0.77	4.09	0.57
2025	Special Purpose Zone	91	471	5.18	0.71	3.13	0.43
2026	National Park Zone	53	124	2.34	0.45	1.72	0.35
2026	Special Purpose Zone	83	248	2.99	0.58	0.94	0.19

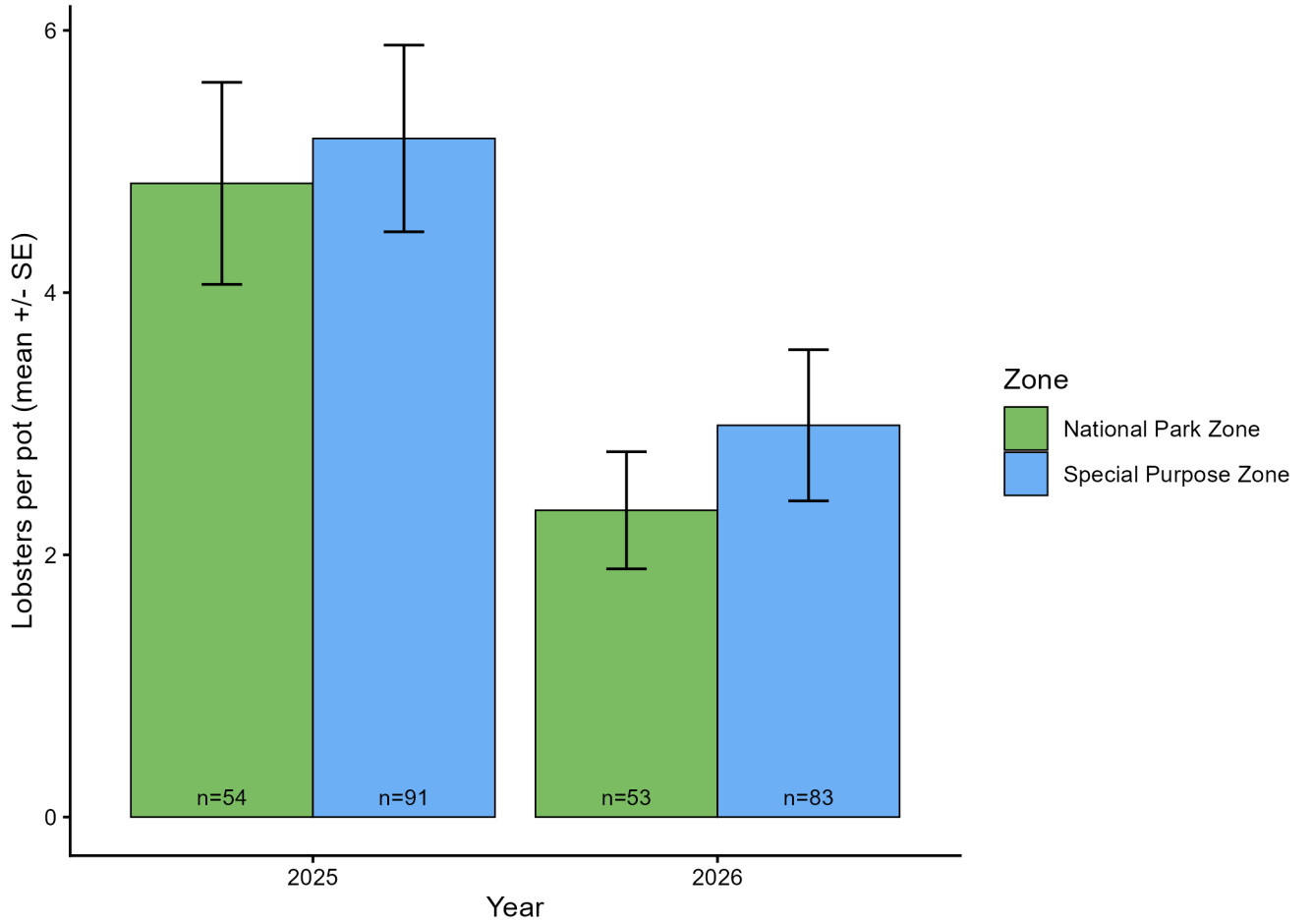


Figure 2: Lobsters per pot by zone and year. Pots that caught nothing are included as zeros.

Western rock lobster abundance per pot

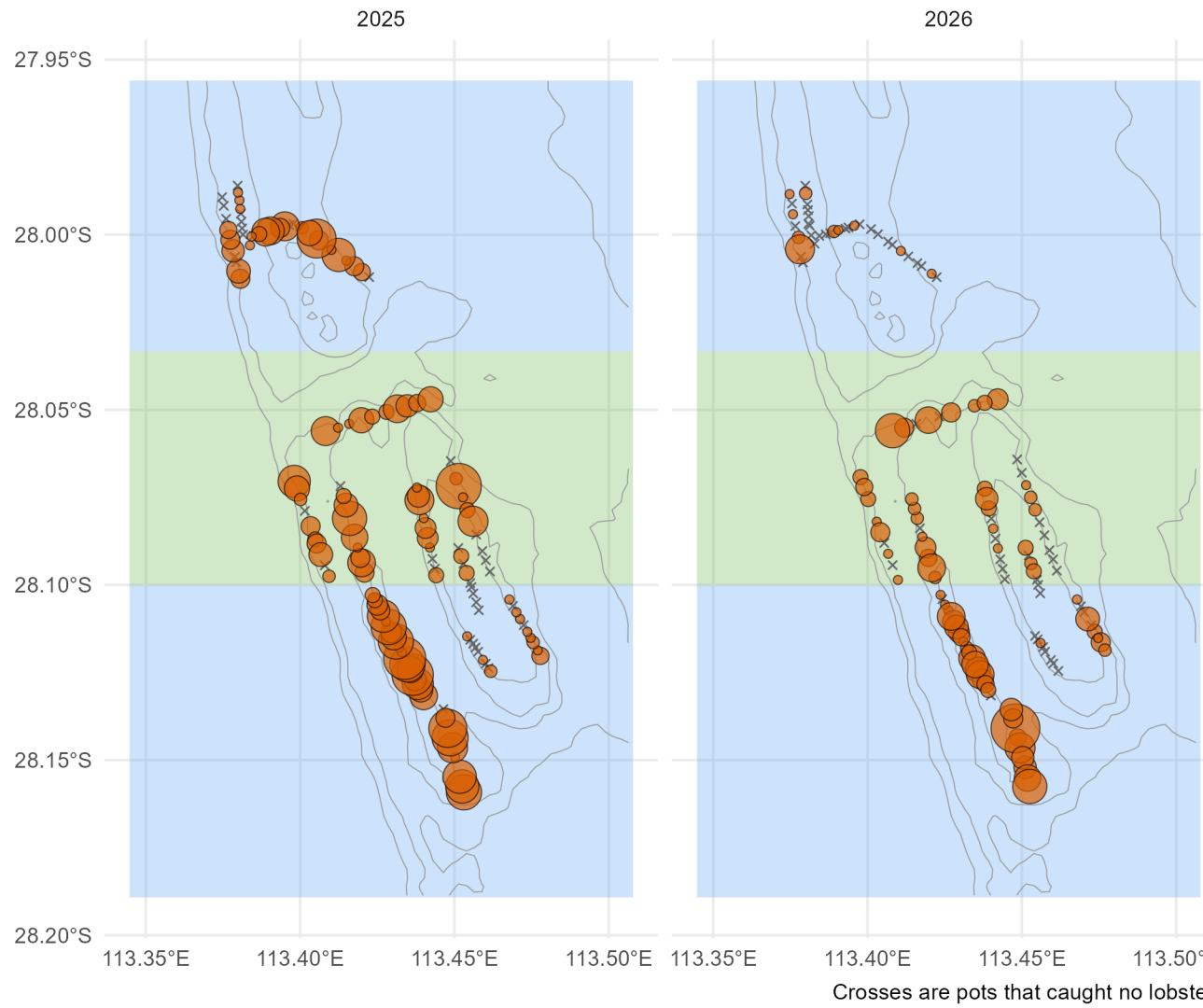


Figure 3: Abundance per pot in each survey year. Crosses mark pots that caught no lobsters.

Legal sized western rock lobster per pot

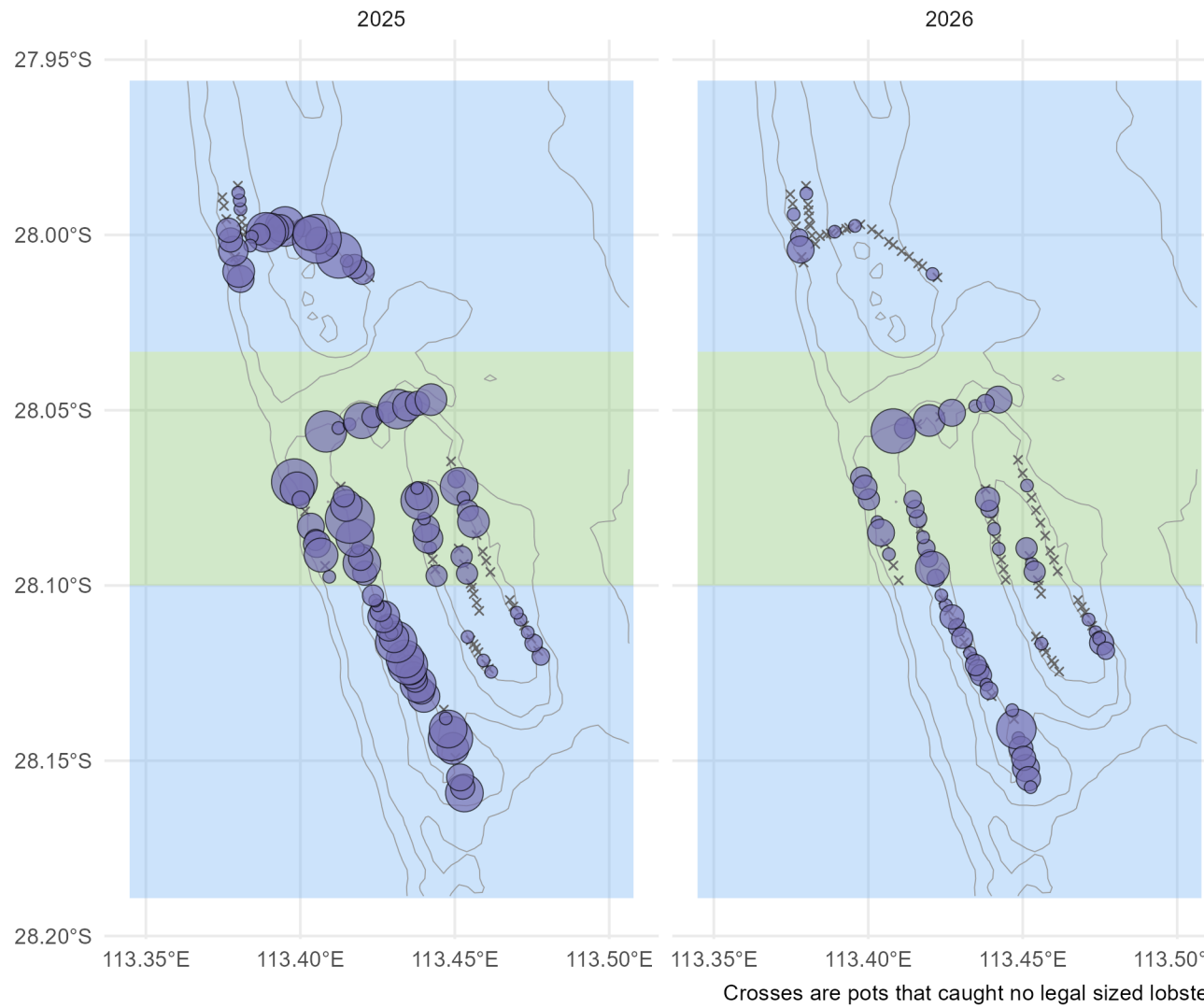


Figure 4: Legal sized lobsters per pot in each survey year

Size structure

The minimum legal carapace length for western rock lobster is 76 mm.

Table 3: Carapace length summary, by year and zone

Year	Zone	n	Mean length (mm)	Median (mm)	Min	Max	% legal
2025	National Park Zone	273	92.7	93	24.0	129.0	85.0
2025	Special Purpose Zone	483	81.6	82	50.0	129.0	60.7
2026	National Park Zone	124	86.6	90	51.0	120.0	73.4
2026	Special Purpose Zone	248	73.0	72	50.3	125.5	31.5

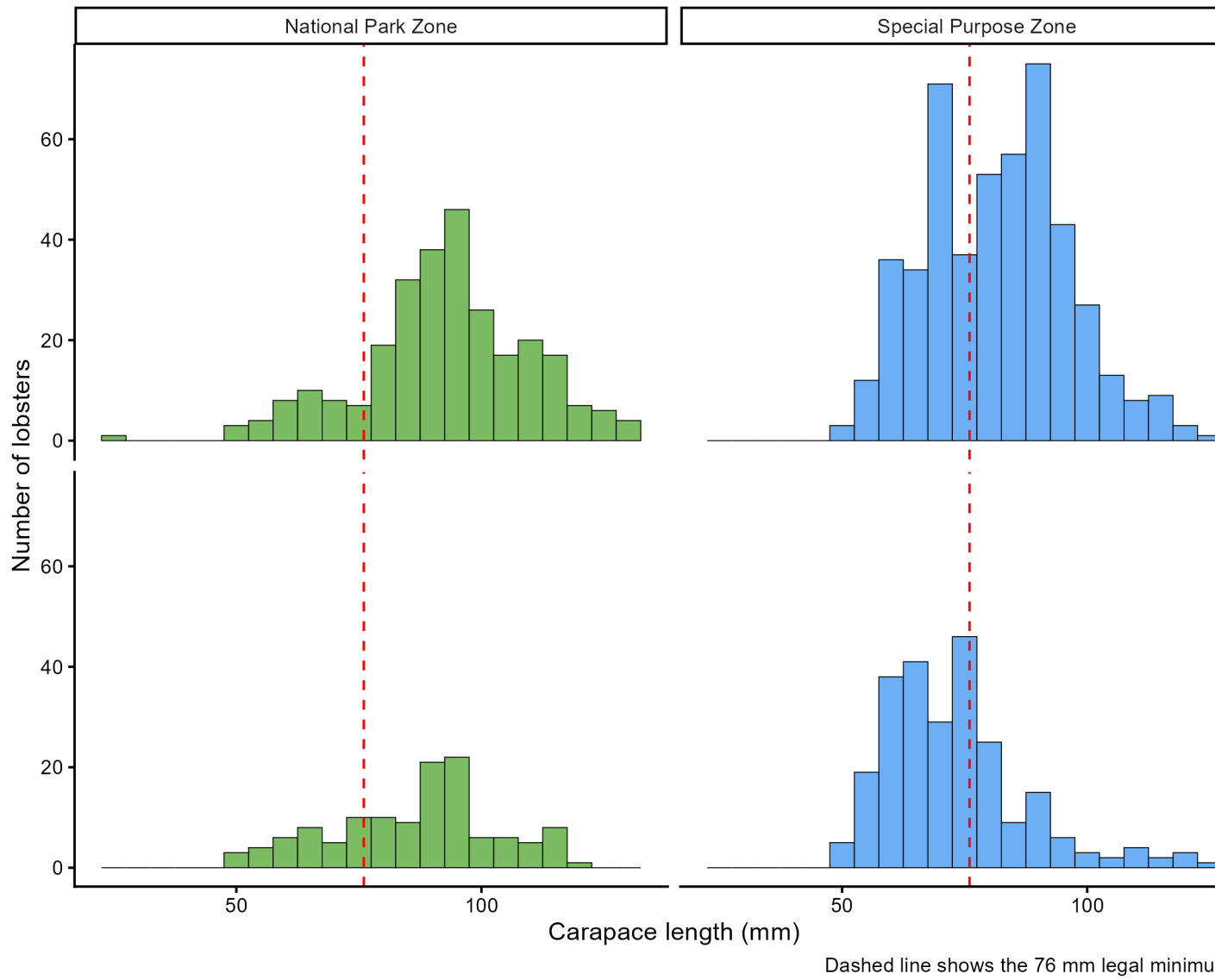


Figure 5: Carapace length distribution by zone and year

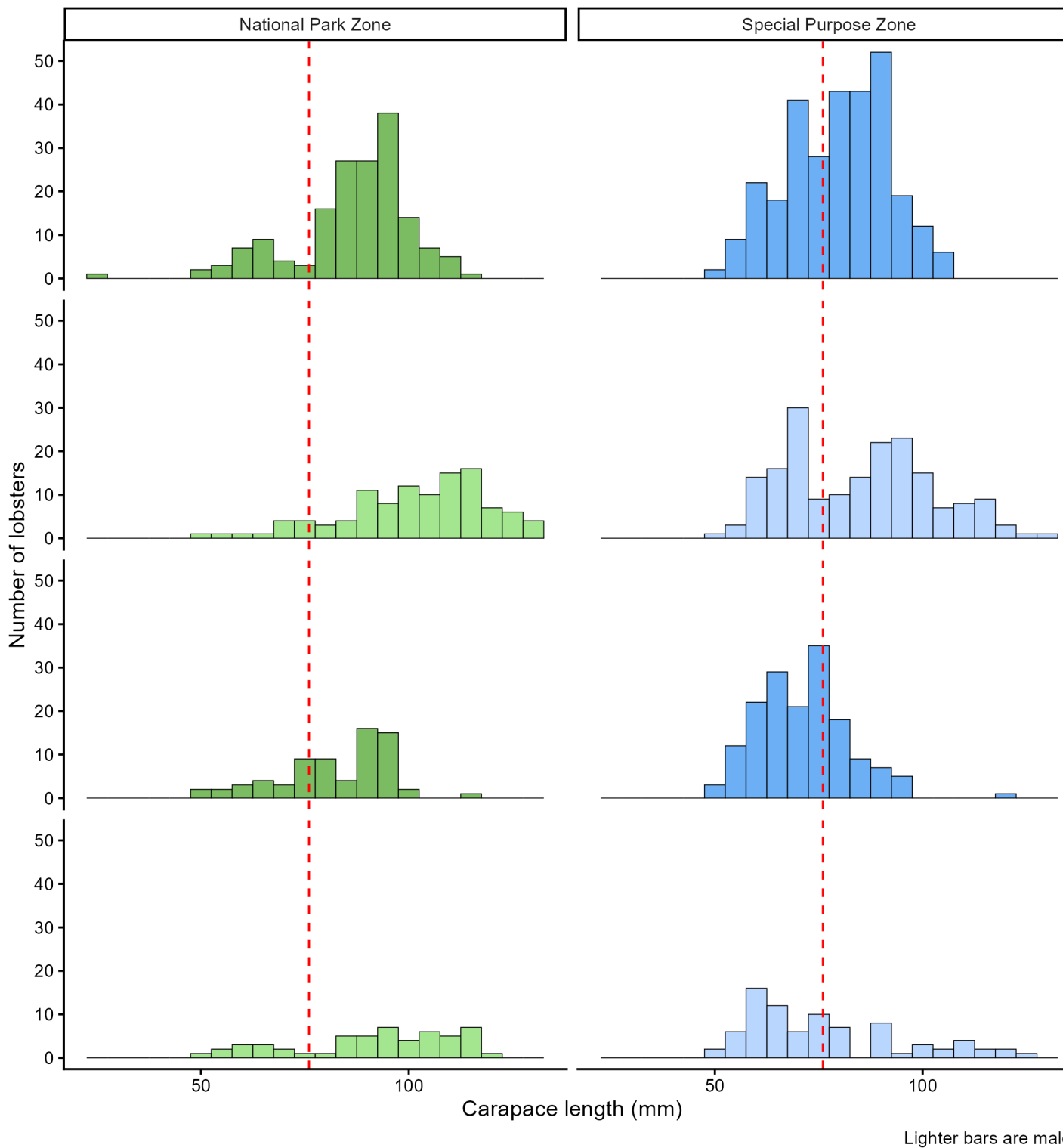


Figure 6: Carapace length distribution by sex. Zone sets the colour and sex sets the shade, with males the lighter bars.

Standardised size distributions

Fewer lobsters were caught in 2026 than 2025, so the histograms above are dominated by differences in abundance. In the two figures below each panel is scaled to sum to 100%, which removes the abundance signal and shows only the shape of the size distribution. Sample sizes are given in each panel, as percentages from a small sample are considerably noisier.

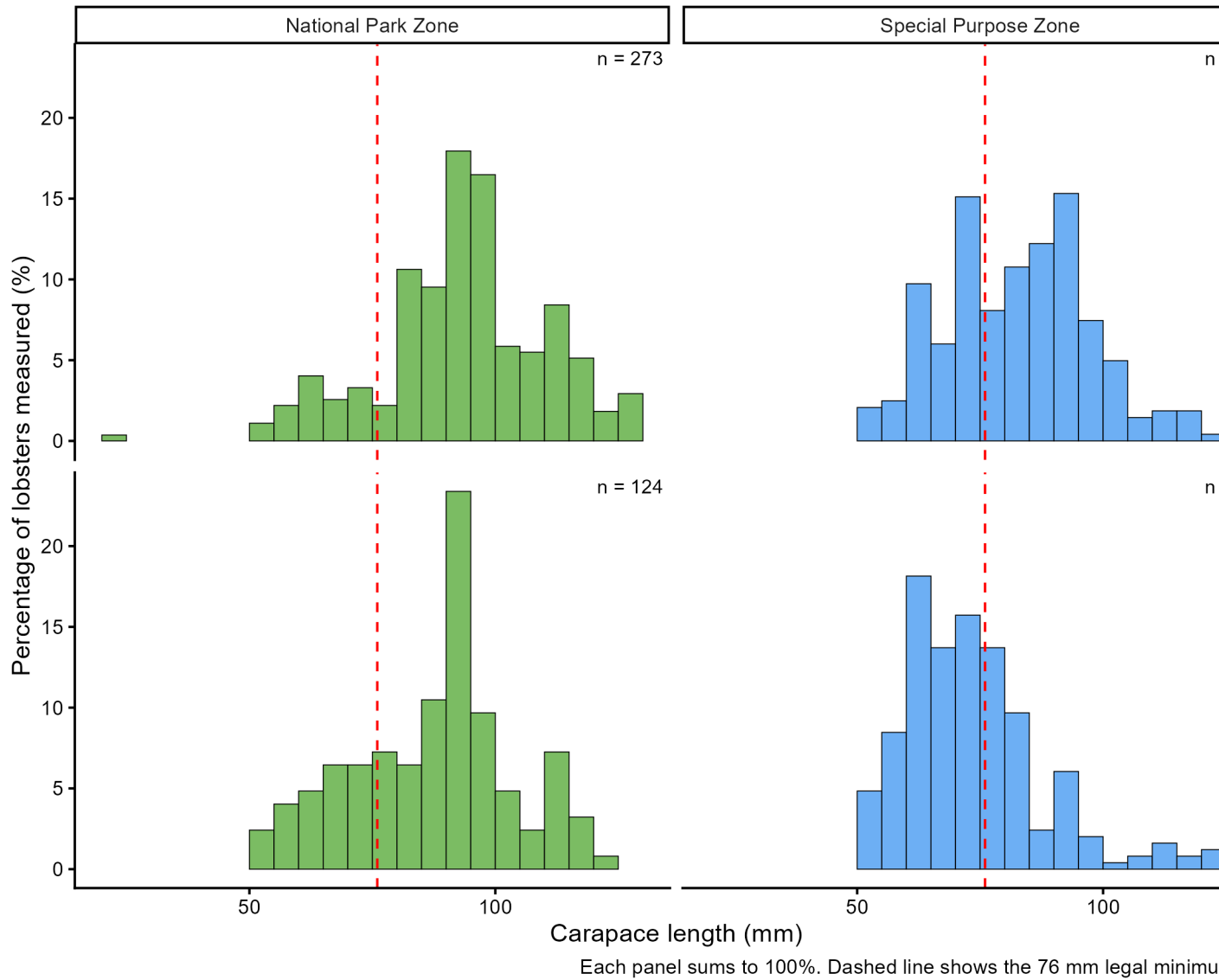


Figure 7: Carapace length distribution by zone and year, standardised so each panel sums to 100%

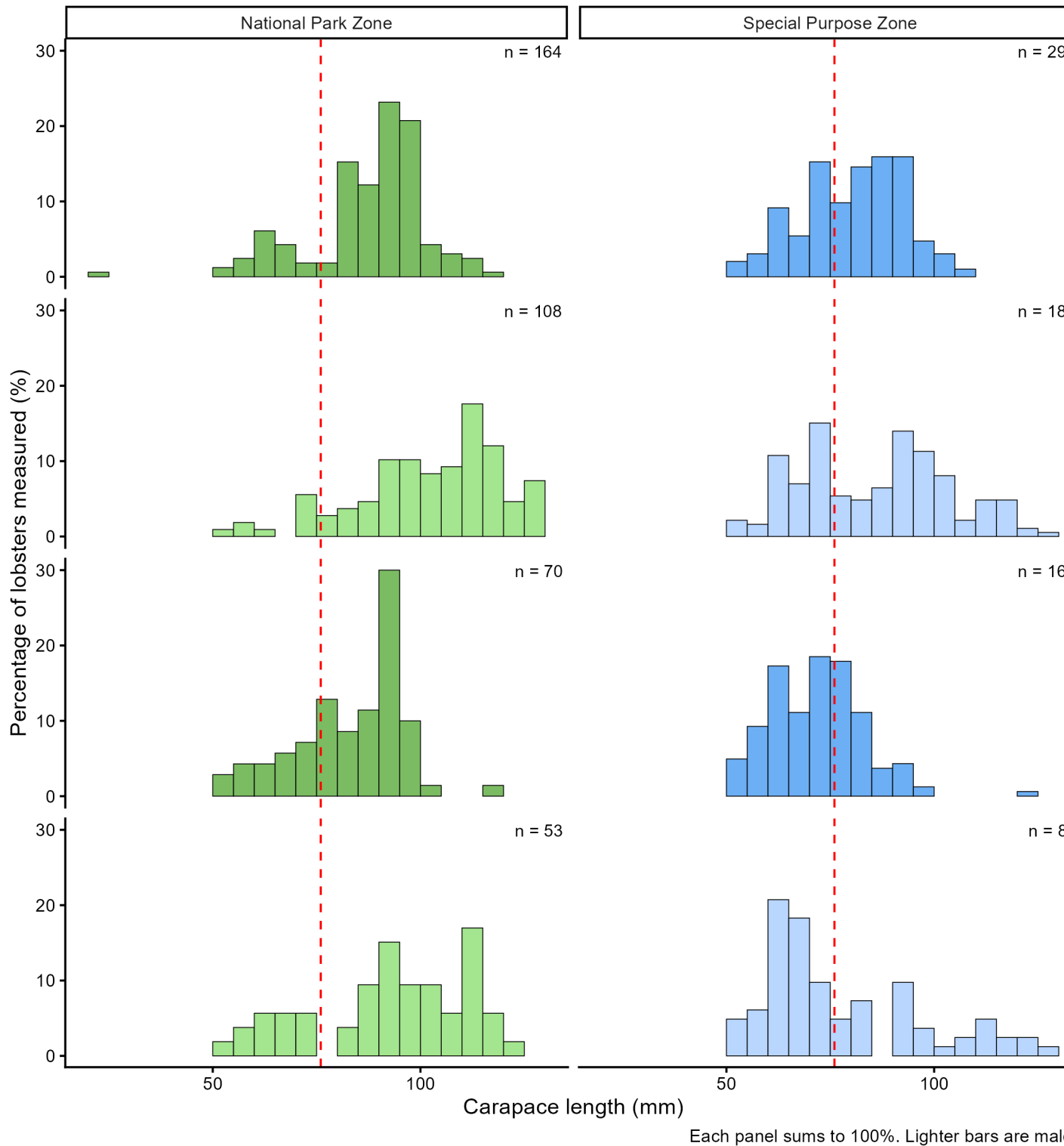


Figure 8: Carapace length distribution by sex, standardised so each panel sums to 100%. Males are the lighter bars.

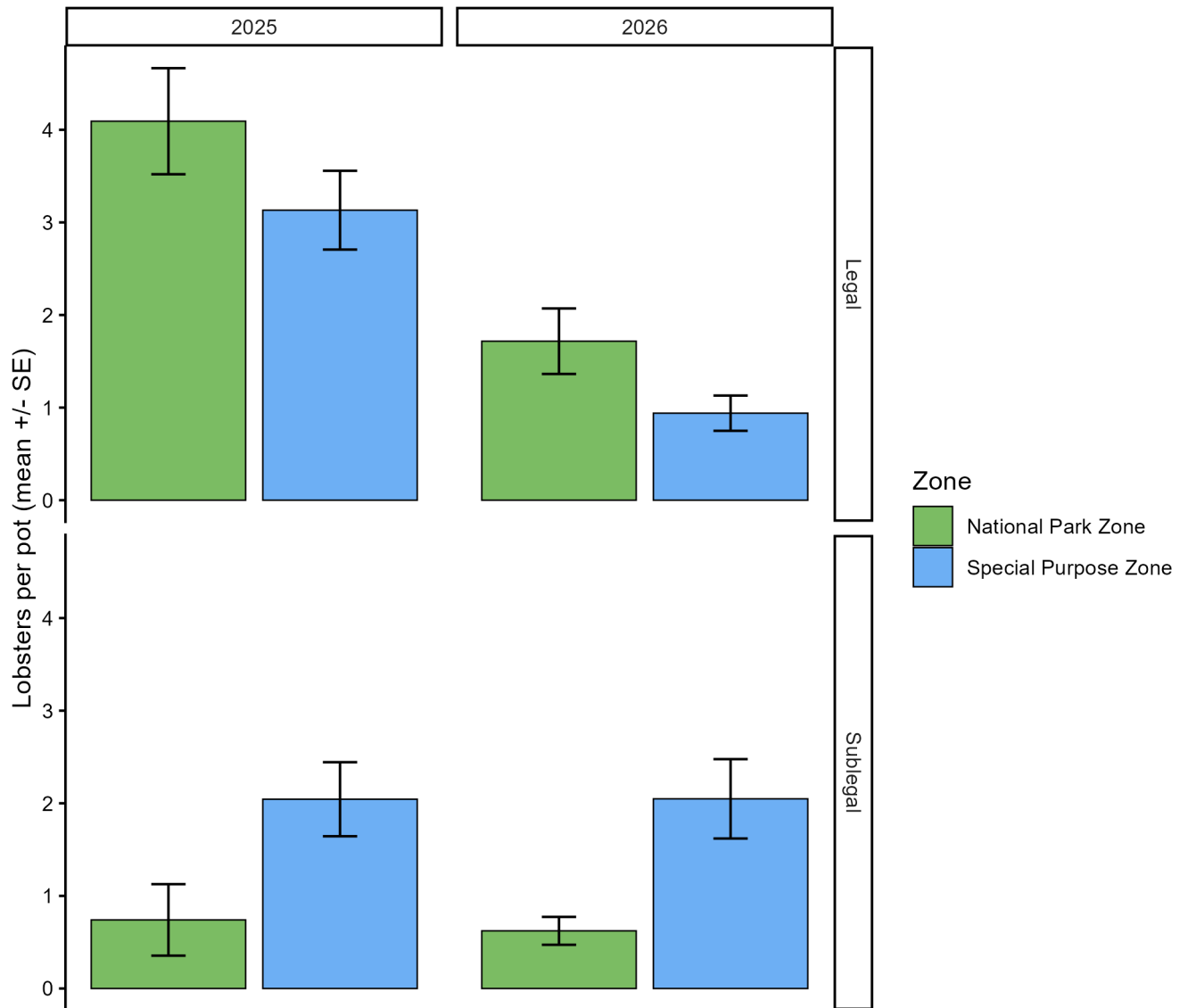


Figure 9: Legal and sublegal catch rates by zone and year

Modelling: catch inside vs outside the National Park Zone

Legal and sublegal catch per pot were modelled separately with a Tweedie generalised linear model (log link), fitted in `glmmTMB`:

```
catch ~ status * year + depth_z
```

where **status** is inside or outside the National Park Zone, **year** is a factor, and **depth_z** is pot depth standardised to zero mean and unit variance. Predicted means and standard errors for each status-by-year combination were obtained with **emmeans**, holding depth at its mean.

The same model was then fitted to legal sized males and legal sized females separately, giving four models in total. Legal females outnumbered legal males roughly three to two, and the two sexes differ in behaviour and in how the fishery takes them, so a change in legal catch can be driven by one sex alone. The sex specific models are subsets of the legal model rather than independent tests of it: legal male and legal female catch together make up the legal catch, apart from 3 legal animals whose sex was not recorded.

Two limitations to read these results against

The random effect for site is not included. The intended specification was `catch ~ status * year + depth_z + (1 | site)`, where site is the string (line) of pots. Neither survey records a string identifier, and the 2026 pot numbering does not follow position along the line, so strings cannot be recovered reliably from the data. Because pots within a string are not independent, the standard errors and p-values below are anti-conservative. The predicted means are much less affected than their uncertainty, so the plots are indicative while the significance tests are provisional.

Status is spatially confounded with survey area. The National Park Zone is a single contiguous block that was fished on its own day in both years. Comparing inside with outside is therefore partly comparing the middle area with the northern and southern areas, and any habitat difference between those areas is not separable from the effect of protection.

Table 4: Predicted lobsters per pot from the Tweedie GLM, by status and year

Size class	Status	Year	Per pot	SE	Lower CL	Upper CL
Legal	Outside NPZ	2025	3.10	0.37	2.45	3.91
Legal	Inside NPZ	2025	3.79	0.55	2.84	5.04
Legal	Outside NPZ	2026	0.95	0.18	0.66	1.37
Legal	Inside NPZ	2026	1.64	0.32	1.12	2.40
Sublegal	Outside NPZ	2025	1.97	0.35	1.38	2.80
Sublegal	Inside NPZ	2025	0.62	0.20	0.32	1.18
Sublegal	Outside NPZ	2026	2.03	0.38	1.41	2.92
Sublegal	Inside NPZ	2026	0.62	0.20	0.32	1.17
Legal male	Outside NPZ	2025	1.18	0.15	0.92	1.52
Legal male	Inside NPZ	2025	1.51	0.23	1.13	2.03
Legal male	Outside NPZ	2026	0.36	0.08	0.23	0.56
Legal male	Inside NPZ	2026	0.75	0.15	0.51	1.12
Legal female	Outside NPZ	2025	1.88	0.25	1.45	2.44
Legal female	Inside NPZ	2025	2.27	0.37	1.65	3.12

Size class	Status	Year	Per pot	SE	Lower CL	Upper CL
Legal female	Outside NPZ	2026	0.58	0.12	0.38	0.88
Legal female	Inside NPZ	2026	0.90	0.20	0.57	1.40

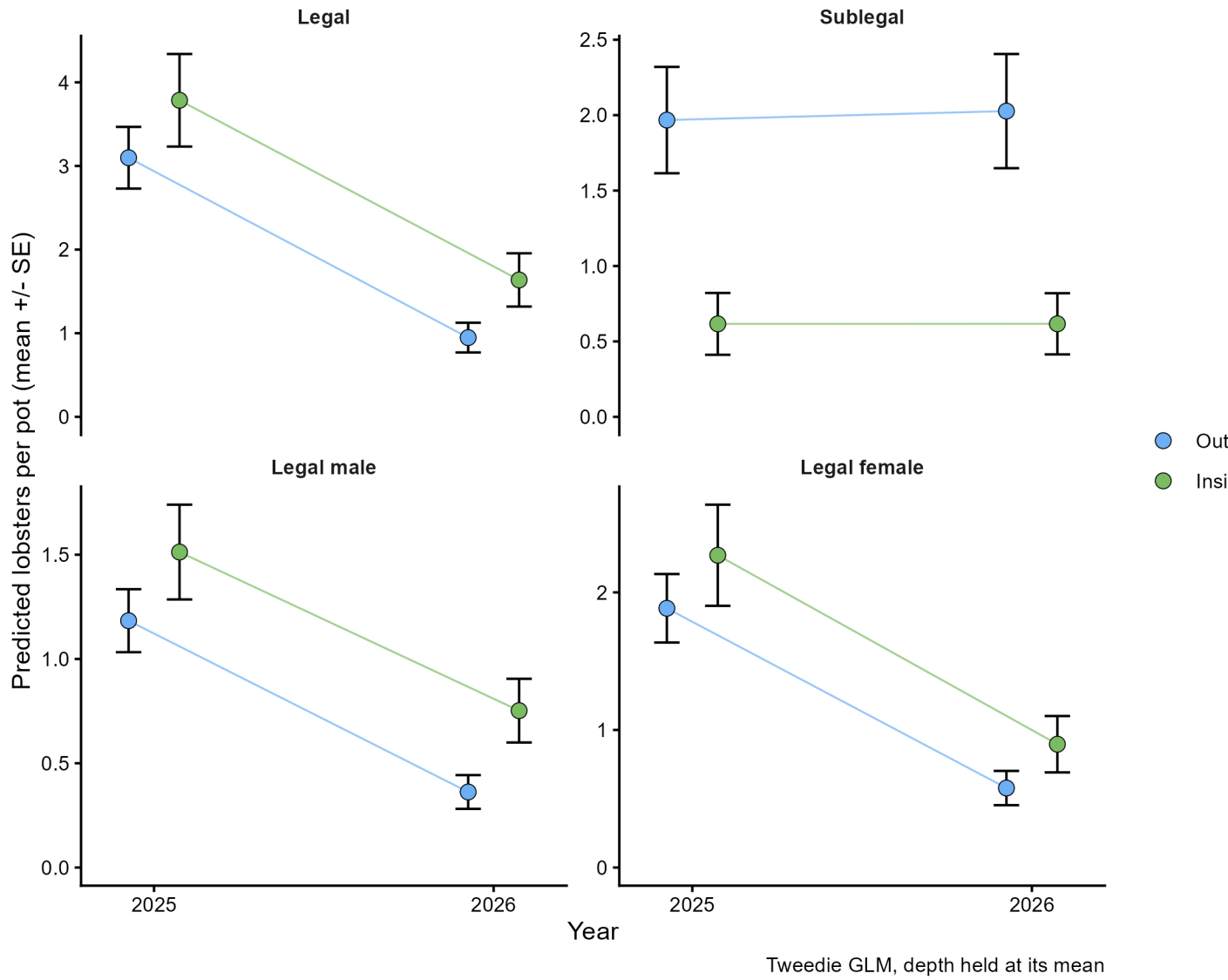


Figure 10: Predicted catch per pot inside and outside the National Park Zone, for each of the four models. Points are model predictions with standard errors, depth held at its mean.

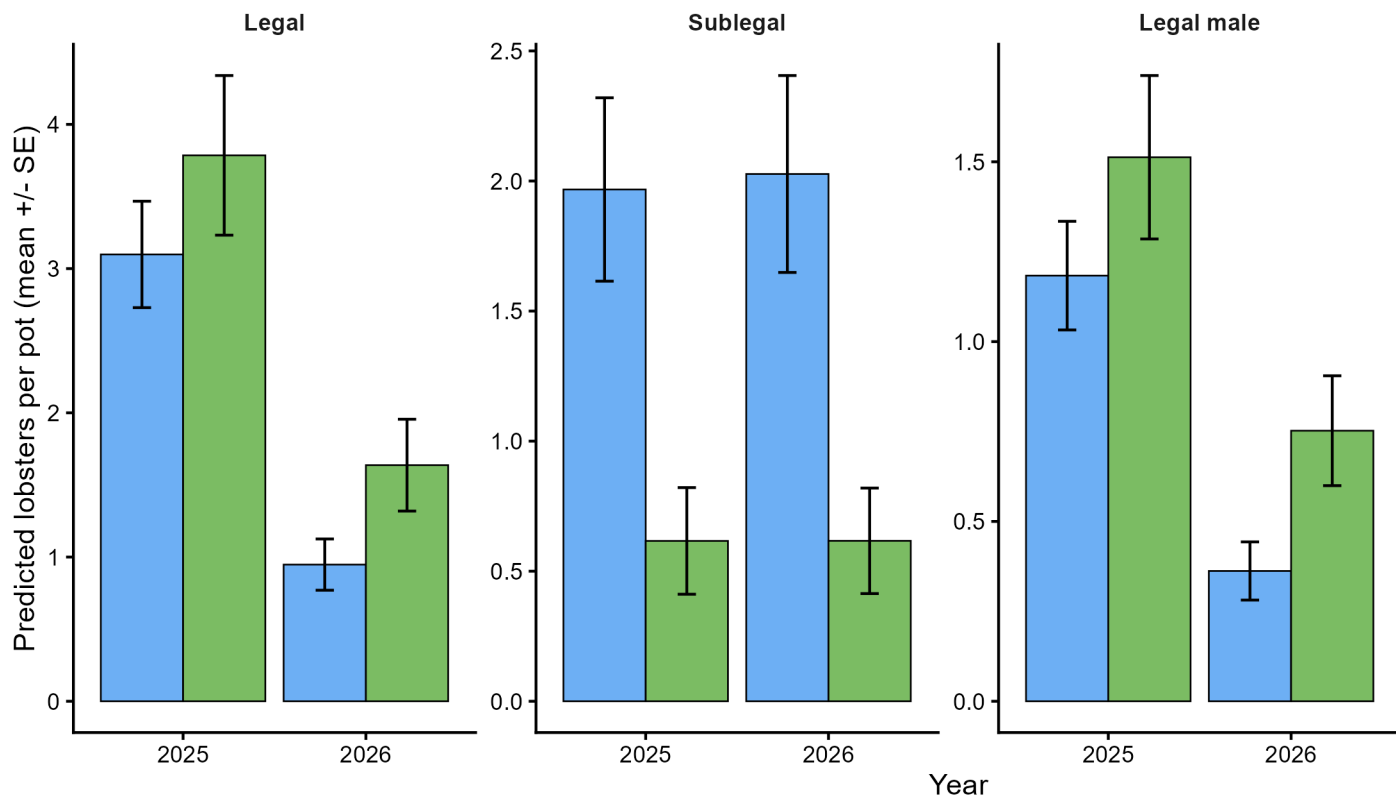


Figure 11: The same predictions shown as bars

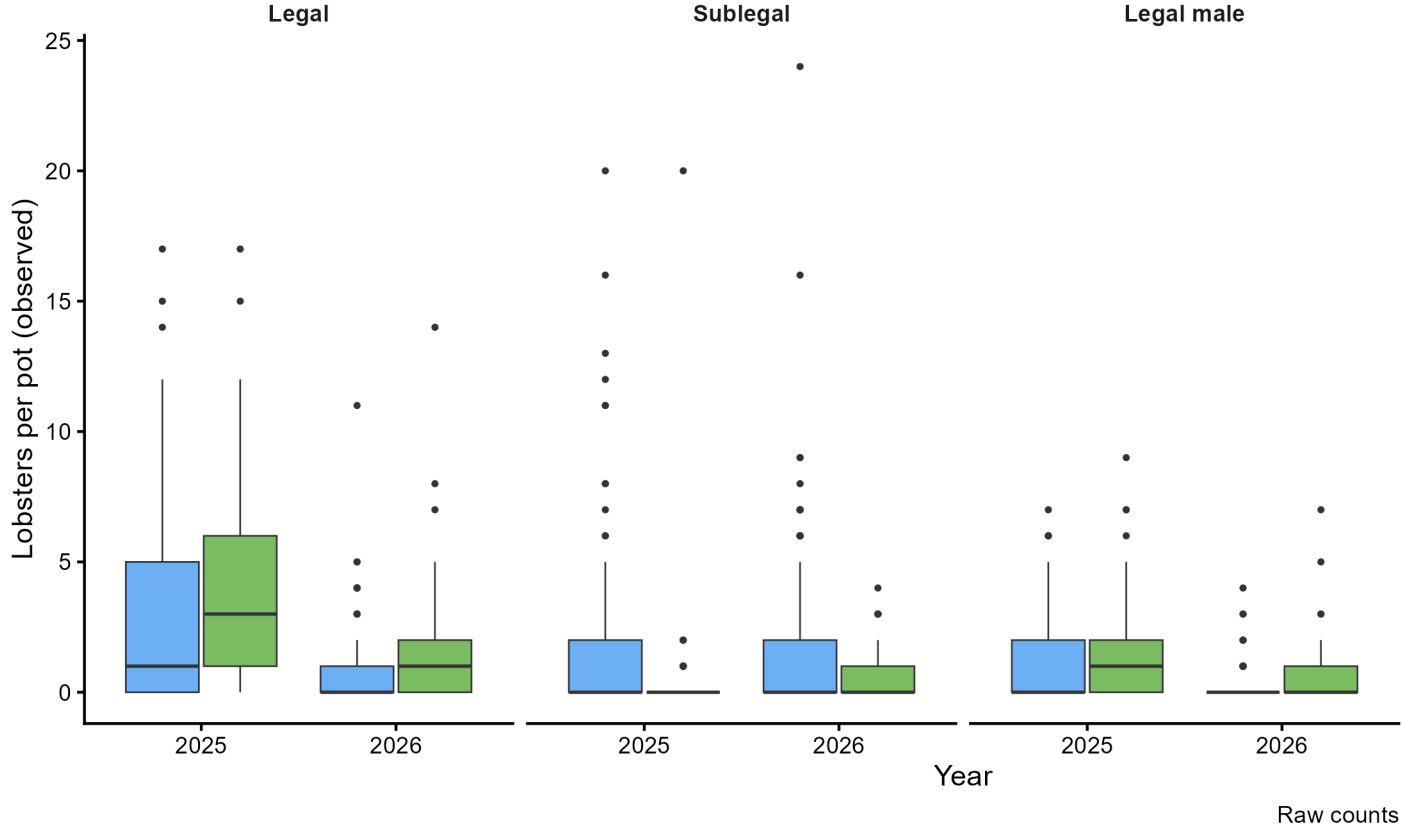


Figure 12: Observed catch per pot, showing the spread behind the modelled means

Table 5: Model coefficients on the log scale. Reference levels are outside the NPZ, 2025.

Size class	Term	Estimate	SE	z	p
Legal	(Intercept)	1.131	0.119	9.50	<0.001
Legal	statusInside NPZ	0.200	0.187	1.07	0.285
Legal	year2026	-1.185	0.222	-5.34	<0.001
Legal	depth_z	0.249	0.084	2.97	0.003
Legal	statusInside NPZ:year2026	0.346	0.328	1.06	0.290
Sublegal	(Intercept)	0.677	0.179	3.78	<0.001
Sublegal	statusInside NPZ	-1.161	0.376	-3.09	0.002
Sublegal	year2026	0.030	0.257	0.12	0.908
Sublegal	depth_z	-0.255	0.134	-1.91	0.057
Sublegal	statusInside NPZ:year2026	-0.029	0.533	-0.05	0.956
Legal male	(Intercept)	0.168	0.128	1.32	0.187
Legal male	statusInside NPZ	0.245	0.196	1.25	0.210

Size class	Term	Estimate	SE	z	p
Legal male	year2026	-1.184	0.257	-4.61	<0.001
Legal male	depth_z	0.218	0.085	2.55	0.011
Legal male	statusInside NPZ:year2026	0.485	0.359	1.35	0.176
Legal female	(Intercept)	0.634	0.132	4.80	<0.001
Legal female	statusInside NPZ	0.186	0.207	0.90	0.369
Legal female	year2026	-1.182	0.252	-4.69	<0.001
Legal female	depth_z	0.268	0.095	2.83	0.005
Legal female	statusInside NPZ:year2026	0.253	0.375	0.68	0.499

Table 6: Likelihood ratio test of the status by year interaction

Size class	Chi squared	df	p
Legal	1.116	1	0.291
Sublegal	0.003	1	0.956
Legal male	1.834	1	0.176
Legal female	0.455	1	0.500

Legal catch fell sharply between years (0.31 times the 2025 rate outside the zone, $p < 0.001$) and was higher inside the National Park Zone in both years, though that difference was not statistically significant on its own ($p = 0.29$). Sublegal catch showed the opposite pattern: substantially lower inside the zone ($p = 0.002$) and unchanged between years. Neither interaction was significant, meaning the difference between zones did not detectably change from 2025 to 2026 — which is expected with only two years of data.

Deeper pots caught more legal lobsters ($p = 0.003$) and slightly fewer sublegal lobsters ($p = 0.057$). Mean depth was almost identical inside and outside the zone (43.9 m and 43.0 m), so depth is not driving the status comparison.

Legal males and females

Both sexes tell the same story as the combined legal model, and neither is driving it on its own. Legal catch fell by close to the same factor between years for males (0.31) and females (0.31), both $p < 0.001$. Both sexes were more abundant inside the National Park Zone in both years, and in neither case was that difference significant on its own (males $p = 0.21$, females $p = 0.37$), nor was either interaction (males $p = 0.18$, females $p = 0.50$). Deeper pots caught more of both sexes (males $p = 0.011$, females $p = 0.005$).

The apparent zone effect is larger for males than females — inside the zone, predicted male catch was 1.28 times the outside rate in 2025 and 2.08 times in 2026, against 1.20 and 1.55 for

females — which is the direction expected if fishing pressure outside the zone falls more heavily on males. With only two years of data, neither the sex difference nor the zone difference itself is statistically supported, and the site caveat above applies equally here.

DHARMA diagnostics were acceptable for three of the four models. The legal male model showed mild overdispersion (dispersion 1.34, $p = 0.032$) with no zero inflation, so its standard errors are likely slightly optimistic on top of the site limitation already noted.

Data notes

- Pots that were lost, had a missing trap door, or were flagged in the field as not countable are excluded from catch rates but retained in the pot table.
- Pots that caught no lobsters are carried through the analysis as zeros.
- 2026 includes six extra pots labelled F3, F3-2 and FIII. These are kept as distinct pots rather than merged with the numbered pots.
- Three 2026 pots lost GPS fix on retrieval; corrected positions were read from the field comments.
- Non-lobster catch from 2026 is written to a separate bycatch file and is not included in these results.
- One 2025 measurement of 24 mm (National Park Zone, pot 2) sits well below the next smallest lobster at 50 mm and is likely a data entry error. It has been retained pending confirmation.